

Jas Shah

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Education

Stony Brook University, New York, USA

Aug 2024 – May 2026

Master of Science in Data Science (Computer Science track)

GPA: 3.73 / 4.00 (Top 10%)

Courses: Statistical Learning, Reinforcement Learning, Big Data, Data Analysis, Visualization, Programming Abstractions

University of Mumbai, Mumbai, India

Aug 2020 – Jun 2024

B.E. in Information Technology

GPA: 9.2 / 10

Courses: Object Oriented Programming, OS, Databases, DSA, Cloud Computing, Distributed Systems, Networking

Technical Skills

AI/ML: Claude API, MCP, LangChain, DSPy, HuggingFace Transformers, PyTorch, scikit-learn, LoRA Fine-Tuning

Languages: Python, C/C++, Java, Go, JavaScript, TypeScript, SQL, Bash

Data & Cloud: AWS (EC2, RDS, S3, IAM), GCP, Databricks, Apache Spark, Kafka, PostgreSQL, Redis, MongoDB

Backend: FastAPI, Spring Boot, REST, GraphQL, gRPC, LevelDB, Docker, Kubernetes, CI/CD, OpenTelemetry, Linux (x86)

Frontend: React, Next.js, Redux Toolkit, Tailwind CSS, Shadcn UI, Jest, Vitest, WCAG 2.1

Experience

Stony Brook University – Software Developer (Research), New York, USA

Aug 2025 – Present

- Partnered with political science researchers and faculty to translate **Fair Representation Act** requirements into a graph-based gluing algorithm, enforcing contiguity, compactness, and proportional seat allocation via **BFS** across all 50 U.S. states.
- Executed 5,000+ **Monte Carlo Markov Chain** ensemble simulations using **MGGG GerryChain** on **Seawulf HPC**, a **Linux x86-based supercomputing cluster**; integrated **Slurm** scheduling to boost throughput by **35%**.

Good Ai – Software Developer Intern, San Francisco, CA (Remote)

Jul 2025 – Jun 2026

- Built and presented product prototypes and demos to company **leadership (CEO, CTO)**, **investors**, and **prospective clients** across **recurring** fundraising cycles, translating in-progress AI features into stakeholder-ready narratives.
- Engineered **30+ reusable React components** with **TypeScript** for AI-powered dashboards, enforcing **WCAG 2.1** accessibility standards and maintaining **95%+ test coverage** via **Vitest**.
- Migrated inter-service event delivery to an **AWS SQS + Kinesis** pipeline with granular **IAM least-privilege** roles across **6 consumers**, sustaining **200K+ events/day** at sub-**150ms** lag and eliminating unauthorized cross-service access.
- Instrumented **FastAPI** microservices with **OpenTelemetry** tracing and **Prometheus + Grafana** dashboards with anomaly alerting across **5+ services**, reducing mean time to detect incidents by **35%**.

Kotton King – Software Developer Intern, Singapore (Remote)

Mar 2024 – Aug 2024

- Presented feature and product walkthroughs to the **CTO** throughout the internship, expanding into **recurring** direct client-facing demos that helped bring in new clients for the platform.
- Built high-performance backend services using **Java (Spring Boot)** and **PostgreSQL**, implementing query optimization and **multithreaded concurrency tuning** to reduce response latency by **40%** while scaling to **10K+ daily requests**.
- Hardened backend APIs against SQL injection and abuse by adding parameterized queries and input sanitization across **8 endpoints**, alongside fixed-window rate limiting to throttle repeated malformed requests.
- Designed a **webhook-driven CI/CD pipeline** using **GitHub Actions**, **Docker**, and **Kubernetes** rolling deployments, reducing deployment time by **20%** and eliminating manual release steps across **3 backend services**.

Projects

PaperMind — LangGraph, LangChain, Pinecone, Claude API, FastAPI, Python

[GitHub](#)

- Built a multi-agent research assistant using **LangGraph** with planner, retriever, and critic agents to answer multi-hop questions over academic corpora, achieving **87% answer relevance** on a 200-question arXiv CS evaluation set.
- Designed an end-to-end RAG pipeline ingesting **3,000+ arXiv PDFs** into **Pinecone** with hybrid retrieval (dense + BM25 reranking), reducing hallucination rate from **31% to 9%** on factual citation queries.
- Implemented a LangGraph critic node that scores retrieved context against the user query before generation and triggers re-retrieval when confidence falls below threshold, cutting low-confidence responses by **43%** across 5 evaluation runs.

DevMind — Claude API, MCP, FastAPI, Redis, React, OpenTelemetry, AWS EC2

[GitHub](#)

- Built an autonomous multi-step code review agent tested across **500+ simulated pull requests**, combining the Claude API with MCP-standardized GitHub integrations and a LangGraph-style agentic loop.
- Cut Claude API token costs by **38%** by implementing **Redis caching** of repeated MCP tool results (file reads, diff fetches) and structured prompt compression, sustaining **sub-2s p95 latency** under concurrent agent runs.
- Designed a self-evaluation loop where the agent critiques its own output against a rubric of **12 code quality dimensions**, including security vulnerability checks, before posting structured review comments to the pull request.

MoviePulse — Python, Scikit-learn, FastAPI, Redis, PostgreSQL

[GitHub](#)

- Built a **hybrid recommendation engine** combining **matrix factorization (ALS)** and **content-based filtering** on the **MovieLens 1M dataset** (1M ratings, 6,040 users, 3,883 movies), achieving an **RMSE of 0.85** on held-out test data, a **24% improvement** over a global-mean baseline predictor.
- Improved **top-10 recommendation precision** by **56%** (**Precision@10 of 0.078** vs **0.050** baseline) by tuning latent factor dimensionality and regularization strength through **5-fold cross-validation** across **56 hyperparameter combinations**.
- Cut **inference latency** from **6.9ms to 0.9ms (7.7x)** by precomputing and caching top-N recommendations in **Redis**, serving predictions through a **FastAPI** endpoint sustaining **1,087 req/s** with **200 concurrent requests** and zero failures.

Achievements

ETHDenver 2025 – 2nd Place — Decentralized audio attribution platform; awarded **\$3,000 USDC**.

[Devfolio](#)

Solved **300+ LeetCode** problems (Hard/Medium) — graphs, dynamic programming, advanced data structures.

[Leetcode](#)